Boolean Expressions

In Python, a Boolean expression is any statement or code segment that evaluates to a Boolean value: either True or False. They form the backbone of decision-making and control flow (like if statements and loops).

Comparison Operators

The most common way to create a Boolean expression is by comparing two values using relational operators given in the following table:

Python Operator	*Mathematics Symbol*	*Name*	*Example* (radius is 5)	*Result*
<	<	less than	radius < 0	False
<=	≤	less than or equal to	radius <= 0	False
>	>	greater than	radius > 0	True
>=	≥	greater than or equal to	radius >= 0	True
==	=	equal to	radius == 0	False
!=	≠	not equal to	radius != 0	True

Comparison operators are used to write Boolean expressions as demonstrated by the following Example

```
1 Male = 29
2 Female = 14
3 print(Male < Female)
4 print(Male <= Female)
5 print(Male > Female)
6 print(Male >= Female)
7 print(Male == Female)
8 print(Male != Female)
```

Ex_0401.py

```
False
False
True
True
False
True
```

Logical Operators (and, or, not)

You can combine multiple Boolean expressions using logical operators (and, or, not) as demonstrated by the following Example

Ex_0402.py

```
age = 25
marital_status = "single"
height = 170
weight = 65
print(age < 30 and  marital_status == "single")
print(age < 30 or  height >= 170)
print(age < 25 and  height >= 170)
print(age > 30 and (not  marital_status == "single"))
print(height >= 170 or weight <= 70)
```

```
True
True
False
False
True
```

Selection Statements

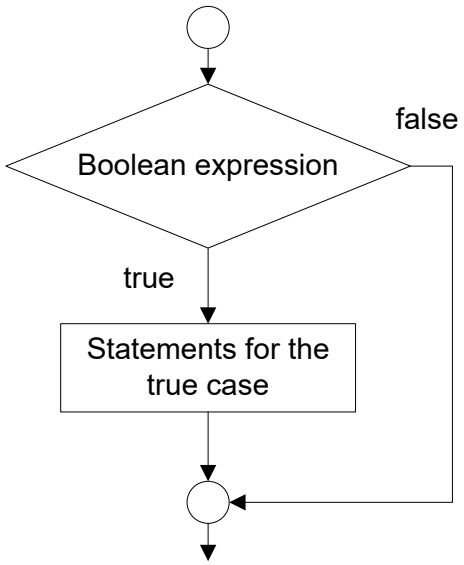
Python provides selection statements that let you choose actions with two or more alternative courses. Selection statements are also known as decision making statements or branching statements. The selection statements are used to select a part of the program to be executed based on a condition. Python provides the following selection statements.

- if statement
- if-else statement
- if-elif statement

if Statement

We use the if statement to test a condition and decide the execution of a block of statements based on that condition result. The if statement checks, the given condition then decides the execution of a block of statements. If it is True, then the block of statements is executed and if it is False, then the block of statements is ignored.

The following flowchart illustrates how Python executes the if statement. If the boolean-expression evaluates to true, the statements in the block are executed, otherwise this block is ignored.



The general syntax of if statement is as follows.

```
if Boolean expression:
    Statement_1
    Statement_2
    Statement_3
    ...
```

When we define an if statement, the block of statements must be specified using **indentation** only. The indentation is a series of white-spaces. Here, the number of white-spaces is variable, but all statements must use the identical number of white-spaces. The following examples demonstrates how to implement if statement.

Suppose we need to check whether an integer number is even or odd. In the following example, the user is prompted to enter an integer number and the program checks whether the number is even odd. If the number is even, the program displays the number is Even. If the number is even, the program displays the number is Odd.

```
1 num = eval(input("Enter an integer number: "))
2
3 if num % 2 == 0:
4     print(num, "is Even")
5
6 if num % 2 != 0:
7     print(num, "is Odd")
```

Ex_0403.py

```
Enter an integer number: 77
77 is Odd
```

```
Enter an integer number: 82
82 is Even
```

In the following example, the student is prompted to enter their total marks for a specific course. The program then checks if the student passed or failed and displays the result.

```
1 marks = eval(input("Enter the sum of marks: "))
2
3 if marks >= 60:
4     print("Congratulations, You Passed the Exam")
5
6 if marks < 60:
7     print("Sorry, You Failed the Exam")
```

Ex_0404.py

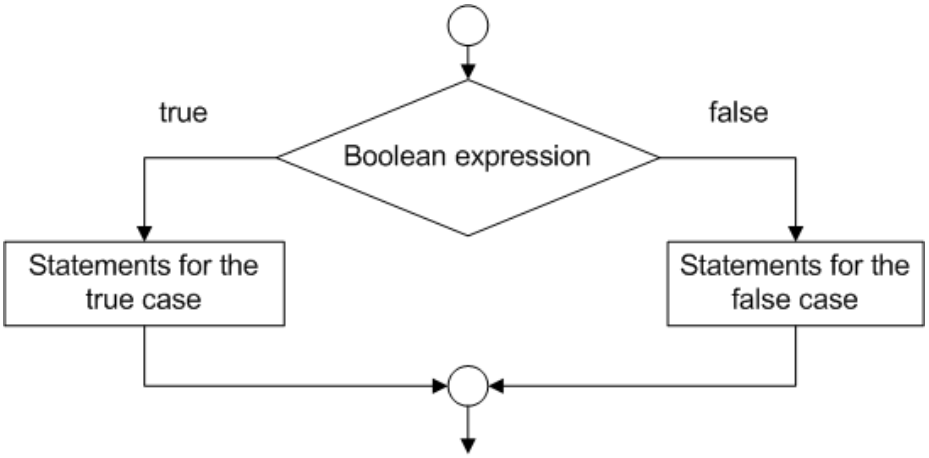
```
Enter the sum of marks: 78
Congratulations, You Passed the Exam
```

```
Enter the sum of marks: 43
Sorry, You Failed the Exam
```

if-else Statement

We use the if-else statement to test a condition and pick the execution of a block of statements out of two blocks based on that condition result. The if-else statement checks the given condition then decides which block of statements to be executed based on the condition result. If the condition is True, then the true block of statements is executed and if it is False, then the false block of statements is executed. The execution flow of if-else statement is as follows.

The following flowchart illustrates how Python executes the if-else statement. If the boolean-expression evaluates to true, the statements in the left block are executed. If the boolean-expression evaluates to false, the statements in the right block are executed.



The general syntax of if-else statement in Python is as follows.

```
if Boolean expression:
        Statement_1
        Statement_2
        Statement_3
        ...
else:
        Statement_4
        Statement_5
        ...
```

In the above syntax, whenever the condition is True, the statements 1 2 and 3 are gets executed. And if the condition is False then the statements 4 and 5 are gets executed.

Let us repeat the last two examples using if-else statement.

```python
num = eval(input("Enter an integer number: "))

if num % 2 == 0:
    print(num, "is Even")
else:
    print(num, "is Odd")
```

Ex_0405.py

```
Enter an integer number: 24
24 is Even
```

```
Enter an integer number: 31
31 is Odd
```

```python
marks = eval(input("Enter the sum of marks: "))

if marks >= 60:
    print("Congratulations, You Passed the Exam")

else:
    print("Sorry, You Failed the Exam")
```

Ex_0406.py

```
Enter the sum of marks: 85
Congratulations, You Passed the Exam
```

```
Enter the sum of marks: 41
Sorry, You Failed the Exam
```

if-elif else statement in Python

When we want to test multiple conditions we use elif statement. The general syntax of if-elif-else statement in Python is as follows.

```
if condition_1:
        Statement_1
        Statement_2
        Statement_3
        ...
elif condition_2:
        Statement_4
        Statement_5
        Statement_6
        ...
else:
        Statement_7
        Statement_8
        ...
```

In the above syntax, whenever the condition_1 is True, the statements 1 2 and 3 are gets executed. If the condition_1 is False and condition_2 is True then the statements 4, 5, and 6 are gets executed. And if condition_1 nad Condition_2 both are False then the statements 7 and 8 are executed.

Let us demonstrate if-elif-else statement by the following Examples.

Example

The phase angle measured at the input terminals of an AC circuit determines whether the circuit is resistive, capacitive, or inductive. Write a Python program that prompts the user to enter the phase angle of the input current. The program should then analyze the input and print the nature of the circuit based on the following criteria:

- Resistive circuit: Phase angle is exactly equal zero
- Capacitive circuit: Phase angle is greater than zero
- Inductive circuit: Phase angle is less than zero

```python
# A program to determine the nature of an AC circuit
theta = eval(input("Enter the phase angle in degrees (-90 to 90): "))
if theta == 0:
    print("Purely resistive circuit")
elif -90 <= theta < 0:
    print("Inductive circuit")
elif 0 < theta <= 90:
    print("Capacitive circuit")
else:
    print("Error input - Invalid phase angle")
```

Ex_0407.py

```
Enter the phase angle in degrees (-90 to 90): 0
Purely resistive circuit
```

```
Enter the phase angle in degrees (-90 to 90): -30
Inductive circuit
```

```
Enter the phase angle in degrees (-90 to 90): 45
Capacitive circuit
```

```
Enter the phase angle in degrees (-90 to 90): 150
Error input - Invalid phase angle
```

Case Study: Computing Body Mass Index (BMI)

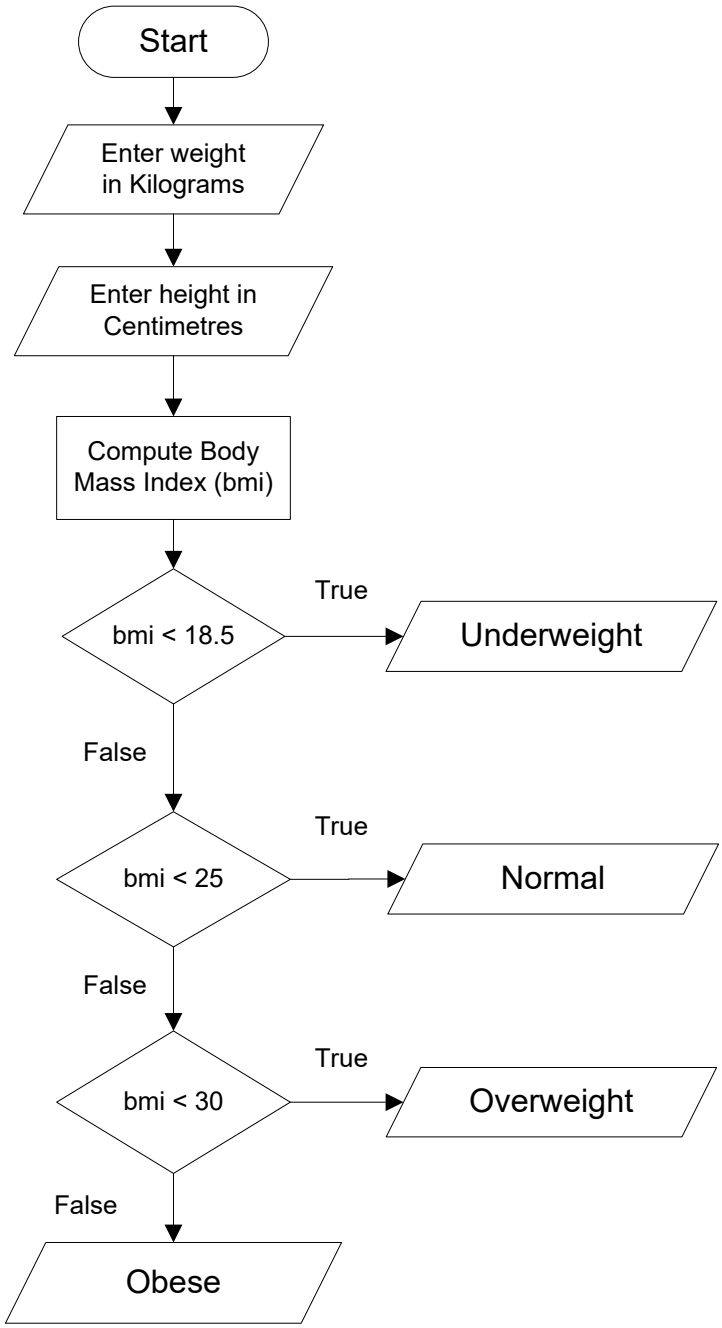
Body Mass Index (BMI) is a screening tool that estimates body fat based on a person's weight and height. It is calculated by dividing weight (in kilograms) by height squared (in meters). That is

$$BMI = \frac{Weight\ in\ Kg}{(Height\ in\ m)^2}$$

Write a Python program that prompts the user to enter a person's weight in kilograms and height in meters. The program calculates the BMI and gives an interpretation for the given data according the following table;

BMI	Interpretation
$BMI < 18.5$	Underweight
$18.5.0 \leq BMI < 25.0$	Normal
$25.0 \leq BMI < 30.0$	Overweight
$30.0 \leq BMI$	Obese

The solution is illustrated first by flowchart and after that it is translated into program code.


Start
Enter weight in Kilograms
Enter height in Centimetres
Compute Body Mass Index (bmi)
bmi < 18.5
True
Underweight
False
bmi < 25
True
Normal
False
bmi < 30
True
Overweight
False
Obese

```python
# Body Mass Index (BMI)
weight = eval(input("Enter the weight in Kgs: "))
height = eval(input("Enter height in Centimetres: "))

# Compute Body Mass Index (BMI)
bmi = weight / (height / 100)**2

# Calculate BMI and display the result
if bmi < 18.5:
    print("BMI = ",format(bmi,"<10.1f"), "Underweight")
elif 18.5 <= bmi < 25.0:
    print("BMI = ",format(bmi,"<10.1f"), "Normal")
elif 25.0 <= bmi < 30.0:
    print("BMI = ",format(bmi,"<10.1f"), "Overweight")
else:
    print("BMI = ",format(bmi,"<10.1f"), "Obese")
```

Ex_0408.py

Example

We need to calculate the total percentage and grade of a student in five courses. Write a Python program to prompt the user to enter the marks of each course, calculate the total percentage and the grade corresponding to the total percentage. Assume that the maximum marks for each course is 100. Grades are determined according to the following table;

Score	Grade Character
$100 \geq total\ percentage \geq 90$	**A**
$90 > total\ percentage \geq 80$	**B**
$80 > total\ percentage \geq 70$	**C**
$70 > total\ percentage \geq 60$	**D**
$60 > total\ percentage$	**F**

Ex_0409.py

```python
# Input marks for 5 courses (out of 100 each)
course_1 = int(input("Enter the marks of course 1: "))
course_2 = int(input("Enter the marks of course 2: "))
course_3 = int(input("Enter the marks of course 3: "))
course_4 = int(input("Enter the marks of course 4: "))
course_5 = int(input("Enter the marks of course 5: "))

# Calculate total and percentage
total_marks = course_1 + course_2 + course_3 + course_4 + course_5
perc = (total_marks / 500) * 100

# Determine grade
if perc >= 90:
    grade = 'A'
elif perc >= 80:
    grade = 'B'
elif perc >= 70:
    grade = 'C'
elif perc >= 60:
    grade = 'D'
else:
    grade = 'F'

print("Total Percentage = ", format(perc,"8.2f"))
print("Grade = " + grade)
```

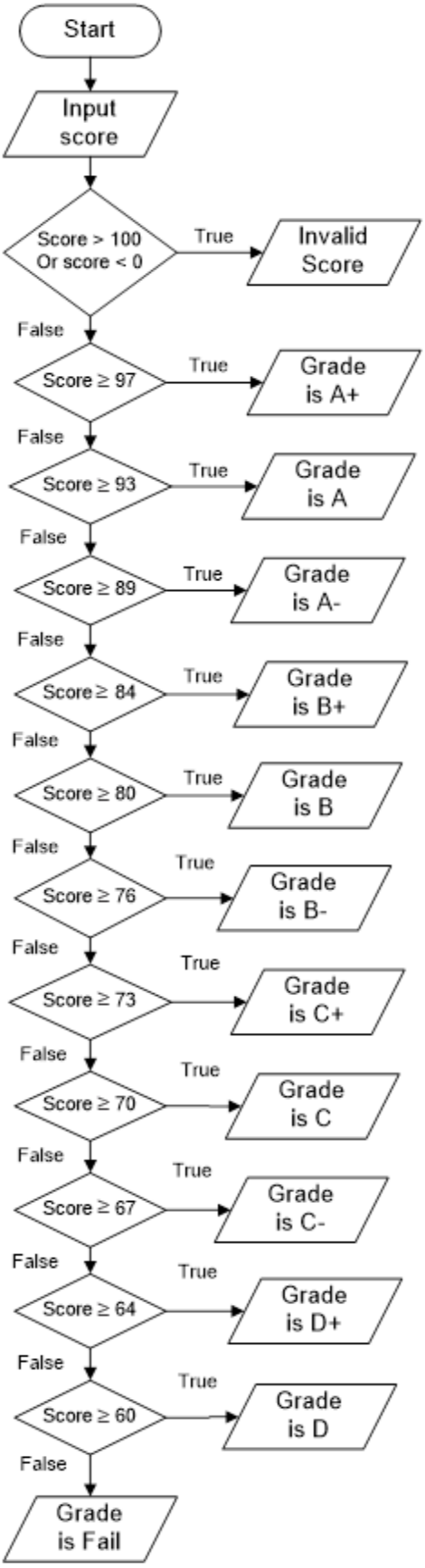
```
Enter the marks of course 1: 87
Enter the marks of course 2: 79
Enter the marks of course 3: 92
Enter the marks of course 4: 85
Enter the marks of course 5: 75
Total Percentage =     83.60
Grade = B
```

Case Study: The Grade Character corresponding to student Score

Design a Python program to accept the student Score in a certain course and obtain the Grade character corresponding to that score according to the following Table:

Score	Grade Character	Score	Grade Character
$100 \geq Score \geq 97$	**A+**	$76 > Score \geq 73$	**C+**
$97 > Score \geq 93$	**A**	$73 > Score \geq 70$	**C**
$93 > Score \geq 89$	**A-**	$70 > Score \geq 67$	**C-**
$89 > Score \geq 84$	**B+**	$67 > Score \geq 64$	**D+**
$84 > Score \geq 80$	**B**	$64 > Score \geq 60$	**D**
$80 > Score \geq 76$	**B-**	$60 > Score$	**F**

The solution is illustrated as follows;


Start
Input score
Score > 100 Or score < 0
True
Invalid Score
False
Score ≥ 97
True
Grade is A+
False
Score ≥ 93
True
Grade is A
False
Score ≥ 89
True
Grade is A-
False
Score ≥ 84
True
Grade is B+
False
Score ≥ 80
True
Grade is B
False
Score ≥ 76
True
Grade is B-
False
Score ≥ 73
True
Grade is C+
False
Score ≥ 70
True
Grade is C
False
Score ≥ 67
True
Grade is C-
False
Score ≥ 64
True
Grade is D+
False
Score ≥ 60
True
Grade is D
False
Grade is Fail

```python
# Prompt the user for the score and convert it to an integer
score = int(input("Enter the score (0:100): "))

# Determine the grade based on the score
if score > 100 or score < 0:
    print(score, " is invalid score")
elif not score:
    print(score, "score should be numeric value")
elif score >= 97:
    print(score,"-->" + " A+")
elif score >= 93:
    print(score,"-->" + " A")
elif score >= 89:
    print(score,"-->" + " A-")
elif score >= 84:
    print(score,"-->" + " B+")
elif score >= 80:
    print(score,"-->" + " B")
elif score >= 76:
    print(score,"-->" + " B-")
elif score >= 73:
    print(score,"-->" + " C+")
elif score >= 70:
    print(score,"-->" + " C")
elif score >= 67:
    print(score,"-->" + " C-")
elif score >= 64:
    print(score,"-->" + " D+")
elif score >= 60:
    print(score,"-->" + " D")
else:
    print(score,"-->" + " F")
```

Ex_0410.py